

# **X-ray Diffraction from Crystals**

**Path difference, structure factor, and systematic absences**

Murali Palla  
BITS Pilani

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# Geometry and assumptions

Let the incident and scattered wavevectors be  $\mathbf{k}_i$  and  $\mathbf{k}_f$ , with  $|\mathbf{k}_i| = |\mathbf{k}_f| = k = 2\pi/\lambda$ . Define the scattering vector

$$\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i.$$

- Elastic scattering:  $|\mathbf{k}_i| = |\mathbf{k}_f|$ .
- A lattice point is  $\mathbf{R}_{mn} = m\mathbf{a}_1 + n\mathbf{a}_2$ .
- The atomic structure factor is taken angle-independent:  
 $f_j(\theta) = f_j$ .

For a symmetric reflection, the angle between the incident beam and the lattice planes is  $\theta$ , hence the angle between the incident and diffracted beams is  $2\theta$ .

# Path difference

For atoms at  $\mathbf{r}$  and  $\mathbf{r} + \mathbf{R}$ , the extra distance travelled by the incident ray is  $-\hat{\mathbf{k}}_i \cdot \mathbf{R}$ , while that on exit is  $+\hat{\mathbf{k}}_f \cdot \mathbf{R}$ . Therefore

$$\Delta(\mathbf{R}) = (\hat{\mathbf{k}}_f - \hat{\mathbf{k}}_i) \cdot \mathbf{R}$$

and the phase difference is

$$\phi(\mathbf{R}) = \frac{2\pi}{\lambda} \Delta(\mathbf{R}) = \mathbf{q} \cdot \mathbf{R}.$$

For  $\mathbf{R}_{mn} = m\mathbf{a}_1 + n\mathbf{a}_2$ ,

$$\phi_{mn} = m\mathbf{q} \cdot \mathbf{a}_1 + n\mathbf{q} \cdot \mathbf{a}_2$$

Constructive interference occurs when  $\phi_{mn}$  is an integer multiple of  $2\pi$  for every lattice translation.

# Laue condition and Bragg form

The intensity is maximized when

$$\mathbf{q} \cdot \mathbf{a}_1 = 2\pi h, \quad \mathbf{q} \cdot \mathbf{a}_2 = 2\pi k,$$

where  $h, k \in \mathbb{Z}$ . In three dimensions this is

$$\mathbf{q} = \mathbf{G}_{hkl} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3,$$

with  $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$ .

$$|\mathbf{G}_{hkl}| = \frac{2\pi}{d_{hkl}}, \quad |\mathbf{q}| = 2k \sin \theta = \frac{4\pi}{\lambda} \sin \theta.$$

Thus

$$\boxed{2d_{hkl} \sin \theta = n\lambda}$$

which is Bragg's law. The integer  $n$  can be absorbed into the Miller indices, so the first-order reciprocal-lattice condition is normally used.

# Structure factor

Let the unit cell contain atoms  $j = 1, \dots, N$ , at fractional coordinates  $(x_j, y_j, z_j)$ , with  $\mathbf{r}_j = x_j \mathbf{a}_1 + y_j \mathbf{a}_2 + z_j \mathbf{a}_3$ . The scattered amplitude is

$$A(\mathbf{q}) = \sum_{\mathbf{R}} \sum_j f_j \exp[i\mathbf{q} \cdot (\mathbf{R} + \mathbf{r}_j)].$$

At a reciprocal-lattice point,  $\exp(i\mathbf{q} \cdot \mathbf{R}) = 1$ , and the unit-cell amplitude is the structure factor

$$F_{hkl} = \sum_{j=1}^N f_j \exp\{2\pi i(hx_j + ky_j + lz_j)\}.$$

The diffracted intensity is

$$I_{hkl} \propto |F_{hkl}|^2.$$

If all atoms are identical with constant  $f$ , then

$$F_{hkl} = f \sum_j e^{2\pi i(hx_j + ky_j + lz_j)}.$$

# Simple cubic (SC)

The conventional SC cell has one lattice point:

$$(0, 0, 0).$$

Therefore

$$F_{hkl} = f, \quad I_{hkl} \propto f^2.$$

## Extinction rule

There are no systematic absences: all integer  $(hkl)$  are allowed, apart from the trivial exclusion  $(000)$  as a Bragg reflection.

# Body-centred cubic (BCC)

Basis positions:

$$(0, 0, 0), \quad \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right).$$

Hence

$$F_{hkl} = f [1 + e^{\pi i(h+k+l)}] = f [1 + (-1)^{h+k+l}].$$

Extinction rule

$h + k + l = 2n$ : allowed;  $h + k + l = 2n + 1$ : absent.

# Face-centred cubic (FCC)

Basis positions:

$$(0, 0, 0), (0, \frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, 0, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}, 0).$$

Thus

$$F_{hkl} = f [1 + (-1)^{k+l} + (-1)^{h+l} + (-1)^{h+k}].$$

## Extinction rule

Reflections occur only when  $h, k, l$  are all even or all odd. Mixed parity gives  $F_{hkl} = 0$ .

For allowed reflections,  $|F| = 4f$ .



# Hexagonal close-packed (HCP)

For the primitive hexagonal cell, use the two-point basis

$$(0, 0, 0), \quad \left(\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right).$$

The basis factor is

$$F_{hkl} = f \left[ 1 + e^{2\pi i(2h+k)/3} e^{\pi i l} \right].$$

Equivalently,

$$F_{hkl} = f \left[ 1 + (-1)^l e^{2\pi i(2h+k)/3} \right].$$

## Extinction rule (ideal HCP basis)

The basis factor vanishes only when  $l$  is odd and  $2h + k \equiv 0 \pmod{3}$ . For even  $l$ , it is nonzero for every  $h, k$ .

# Diamond cubic (DC)

Diamond cubic is an FCC lattice plus the basis translation

$$\mathbf{t} = \left( \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right).$$

Its structure factor factors as

$$F_{hkl} = f S_{\text{FCC}}(hkl) \left[ 1 + e^{\frac{i\pi}{2}(h+k+l)} \right],$$

where  $S_{\text{FCC}} = 1 + (-1)^{k+l} + (-1)^{h+l} + (-1)^{h+k}$ .

## Extinction rule

First require FCC parity:  $h, k, l$  all even or all odd. If all even, reflections with  $h + k + l \equiv 2 \pmod{4}$  are absent. If all odd, the diamond factor is nonzero; hence all-odd reflections are allowed.

Examples: Si allows 111, 220, and 400; 222 is absent.

# One-page rule sheet

| Lattice       | Systematic condition for nonzero $F_{hkl}$                                    |
|---------------|-------------------------------------------------------------------------------|
| SC            | All $hkl$ allowed.                                                            |
| BCC           | $h + k + l$ even.                                                             |
| FCC           | $h, k, l$ all even or all odd.                                                |
| HCP basis     | Nonzero except when $l$ is odd and $2h + k \equiv 0 \pmod{3}$ .               |
| Diamond cubic | FCC condition plus, for all-even indices, $h + k + l \not\equiv 2 \pmod{4}$ . |

## Important distinction

These are systematic absences caused by translational symmetry. Additional zeros can arise from the actual atomic form factors, multiple chemical species, disorder, thermal motion, or non-ideal atomic positions.

# Conventions and checks

- A different origin changes  $F_{hkl}$  by an overall phase, not the intensity  $|F_{hkl}|^2$ .
- HCP extinction statements depend on the chosen hexagonal basis and on whether one is describing the ideal two-atom basis or a larger conventional cell.
- For a multi-element crystal, replace the common  $f$  by the appropriate  $f_j$  in the general structure-factor sum.
- The angle-independent-atom assumption isolates geometric interference; real X-ray form factors generally depend on angle through  $\sin \theta / \lambda$ .

# BCC selection rules

| Plane      | $N$      | Plane      | $N$       | Plane           | $N$       |
|------------|----------|------------|-----------|-----------------|-----------|
| 100        | 1        | 221        | 9         | <b>411, 330</b> | <b>18</b> |
| <b>110</b> | <b>2</b> | <b>310</b> | <b>10</b> | 331             | 19        |
| 111        | 3        | 311        | 11        | <b>420</b>      | <b>20</b> |
| <b>200</b> | <b>4</b> | <b>222</b> | <b>12</b> | 421             | 21        |
| 210        | 5        | 320        | 13        | <b>332</b>      | <b>22</b> |
| <b>211</b> | <b>6</b> | <b>321</b> | <b>14</b> | <b>422</b>      | <b>24</b> |
| <b>220</b> | <b>8</b> | <b>400</b> | <b>16</b> | <b>431</b>      | <b>26</b> |
|            |          | 410        | 17        | 432             | 29        |

## Allowed reflections shown in orange

For BCC,  $h + k + l$  must be even; equivalently, the allowed values of  $N = h^2 + k^2 + l^2$  in this range are even.

# FCC selection rules

| Plane      | $N$      | Plane      | $N$       | Plane      | $N$       |
|------------|----------|------------|-----------|------------|-----------|
| 100        | 1        | 221        | 9         | 411, 330   | 18        |
| 110        | 2        | 310        | 10        | <b>331</b> | <b>19</b> |
| <b>111</b> | <b>3</b> | <b>311</b> | <b>11</b> | <b>420</b> | <b>20</b> |
| <b>200</b> | <b>4</b> | <b>222</b> | <b>12</b> | 421        | 21        |
| 210        | 5        | 320        | 13        | 332        | 22        |
| 211        | 6        | 321        | 14        | <b>422</b> | <b>24</b> |
| <b>220</b> | <b>8</b> | <b>400</b> | <b>16</b> | 431        | 26        |
|            |          | 410        | 17        | 432        | 29        |

Allowed reflections shown in orange

For FCC,  $h, k, l$  must be all even or all odd; mixed parity is systematically absent.

# Diamond cubic selection rules

| Plane      | <i>N</i> | Plane      | <i>N</i>  | Plane      | <i>N</i>  |
|------------|----------|------------|-----------|------------|-----------|
| 100        | 1        | 221        | 9         | 411, 330   | 18        |
| 110        | 2        | 310        | 10        | <b>331</b> | <b>19</b> |
| <b>111</b> | <b>3</b> | <b>311</b> | <b>11</b> | 420        | 20        |
| 200        | 4        | 222        | 12        | 421        | 21        |
| 210        | 5        | 320        | 13        | 332        | 22        |
| 211        | 6        | 321        | 14        | <b>422</b> | <b>24</b> |
| <b>220</b> | <b>8</b> | <b>400</b> | <b>16</b> | 431        | 26        |
|            |          | 410        | 17        | 432        | 29        |

## Allowed reflections shown in orange

Diamond cubic first obeys FCC parity; all-even reflections with  $h + k + l \equiv 2 \pmod{4}$  are additionally absent.

# $\sin^2 \theta$ ratios for allowed reflections

**SC**     1 : 2 : 3 : 4 : 5 : 6 : 8 : ...

**BCC**   1 : 2 : 3 : 4 : 5 : 6 : 7 : ...

**FCC**   3 : 4 : 8 : 11 : 12 : ...

**DC**     3 : 8 : 11 : 16 : ...

## How to use the sequence

For a cubic crystal,  $\sin^2 \theta = (\lambda^2/4a^2)N$ , where  $N = h^2 + k^2 + l^2$ . Each row is shown in its conventional integer scale; the BCC row is the allowed  $N$  sequence divided by 2.



# Examples

1. The  $\sin^2 \theta$  values calculated from the peak positions of an x-ray powder diffraction of an element were as follows: 0.120, 0.238, 0.357, 0.475, 0.593, 0.711, 0.83. Copper  $K\alpha$  radiation was used. Determine the crystal structure and the lattice parameter.
2. Only two reflections are observed on the powder pattern of a cubic crystal at Bragg angles of  $34.06^\circ$  and  $65.51^\circ$ . Chromium  $K\alpha$  radiation is used. Determine the crystal structure (if possible) and the lattice parameter. Show that the third and higher reflections cannot occur.

# Solution

Given  $\sin^2 \theta = 0.120, 0.238, 0.357, 0.475, 0.593, 0.711, 0.830$ ,

$$\frac{\sin^2 \theta_i}{0.120} = 1.000, 1.983, 2.975, 3.958, 4.942, 5.925, 6.917, \\ \approx 1 : 2 : 3 : 4 : 5 : 6 : 7.$$

The near-integer sequence includes 7. SC skips  $N = 7$ , while FCC and DC do not begin 1 : 2. Thus the crystal is BCC, with  $N = 2, 4, 6, 8, 10, 12, 14$ , indexed as

$$(110), (200), (211), (220), (310), (222), (321).$$

Using Cu  $K\alpha$  radiation,  $\lambda = 1.5406 \text{ \AA}$ ,

$$a_i = \frac{\lambda \sqrt{N_i}}{2 \sqrt{\sin^2 \theta_i}} = 3.145, 3.158, 3.158, 3.161, 3.163, 3.165, 3.164 \text{ \AA}.$$

## Result

The crystal is **BCC**, with  $\bar{a} = 3.159 \text{ \AA} \approx \boxed{3.16 \text{ \AA}}$ .

For  $\theta_1 = 34.06^\circ$ ,  $\theta_2 = 65.51^\circ$ , and Cr  $K\alpha$  radiation ( $\lambda = 2.29 \text{ \AA}$ ),

$$\sin^2 \theta_1 = 0.3137, \quad \sin^2 \theta_2 = 0.8282, \quad \frac{\sin^2 \theta_2}{\sin^2 \theta_1} = 2.640 \approx \frac{8}{3}.$$

The ratio matches the first two diamond-cubic lines,  $N = 3$  (111) and  $N = 8$  (220). Assuming all lower-angle allowed lines would be observed, the missing FCC  $N = 4$  (200) line makes the pattern consistent with **diamond cubic**.

$$a_{111} = \frac{2.29\sqrt{3}}{2 \sin 34.06^\circ} = 3.541 \text{ \AA}, \quad a_{220} = \frac{2.29\sqrt{8}}{2 \sin 65.51^\circ} = 3.559 \text{ \AA}.$$

The next allowed DC line is  $N = 11$  (311). Relative to the second line,

$$\sin^2 \theta_3 = \frac{11}{8} \sin^2 65.51^\circ = 1.139 > 1.$$

## Result

$a \approx \boxed{3.55 \text{ \AA}}$ . Since  $\sin^2 \theta_3 > 1$ , the (311) line and every higher line are impossible.

# Thank You

Questions?